

A proof by induction

Farah Aziz | Fictional sample student | Question 1

Claim: $1 + 3 + \dots + (2n - 1) = n^2$ for every integer $n \geq 1$.

Base case

For $n = 1$, the left side is 1 and the right side is $1^2 = 1$. So the statement is true when $n = 1$.

Induction hypothesis

Let $k \geq 1$ be an arbitrary integer. I assume that $1 + 3 + \dots + (2k - 1) = k^2$.

Induction step

For $k + 1$, I add one more odd term to the sum. Using my hypothesis, I get:

$$\begin{aligned} &1 + 3 + \dots + (2k - 1) + (2k + 1) \\ &= k^2 + (2k + 1) \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2 \end{aligned}$$

Conclusion

Therefore the statement is true for $k + 1$ whenever it is true for k . By mathematical induction, $1 + 3 + \dots + (2n - 1) = n^2$ for every integer $n \geq 1$.